# Latent-Space Prediction Error and Fuzzy Risk-Scaled Recovery for Robotic Manipulation

Thanh-Hai Tran [1,2] and Xuan-Bach Le [1,2*]

[1] Faculty of Computer Science and Engineering, Ho Chi Minh City University of Technology (HCMUT), Ho Chi Minh City, Vietnam
[2] Vietnam National University Ho Chi Minh City, Ho Chi Minh City, Vietnam
{tthai.sdh241,lexuanbach}@hcmut.edu.vn

**Abstract.** Industrial pick-and-place lines demand near-zero failure tolerance, yet many safety layers still wait for a fault to manifest before they trigger an indiscriminate Emergency Stop. This work asks whether latent prediction error can instead serve as a graded recovery signal. The proposed closed loop combines a frozen DINOv2 encoder, an RSSM whose KL-divergence prediction error gives an online risk reading, an ANFIS risk mapper, and a controller that applies bounded trajectory corrections scaled by that risk coefficient. In Webots with a UR5e arm and condition-based TRANSIT dynamics, this work evaluates four recoverable motion disturbances ($N = 40$ episodes per mode per scenario) against a binary fail-safe halt (OFF), treating two semantic or actuation faults as outside joint-space recovery. RSR gains concentrate on Jerky (+25.0 pp) and Speed (+42.5 pp, non-overlapping Wilson 95% CIs); smoothing gains instead concentrate on Vibration, Jerky, and Pause, reducing mean joint deviation by 15.4–44.8% (Pause peaks at 44.8%), while Speed's smoothing stays flat under the same joint-velocity cap. OFF/ON is a deployed-policy comparison. CPU-only latency is P95 = 34.24 ms, which sits 65.8% below the 100 ms design budget.

**Keywords:** Anomaly detection · World models · Fuzzy risk · Risk-aware control · Robotics

## 1 Introduction

Industrial robotic arms in pick-and-place lines operate under near-zero failure tolerance, yet most safety layers remain reactive: they wait for a fault to become visible and then halt the task outright. This response protects the cell, but it turns recoverable disturbances into unplanned downtime. Predictive visual models offer a more graceful alternative, but video-diffusion approaches can hallucinate future frames and typically need seconds per clip on GPU hardware, far too slow for real-time control. Expressive latent predictors such as V-JEPA 2 [1] and DreamerV3 [2] avoid hallucination, but they still lack the safety logic and runtime recovery policy needed to translate prediction error into action.

* Corresponding author

This work structures the closed-loop system around three goals. **(G1) Online Risk Detection**: the system reads latent dynamics during TRANSIT rather than pixel reconstruction, which removes any dependence on generated visual predictions. **(G2) Semantic Risk Quantification**: raw prediction error is converted into graduated, human-readable risk levels (Safe/Medium/High/Critical) through fuzzy logic rather than a single binary threshold, following the long-standing fuzzy-set and Takagi–Sugeno modelling tradition [3,4]. **(G3) Risk-Aware Adaptive Control**: instead of halting on the first alarm, the controller continues through moderate risk and issues bounded corrective actions scaled by the risk coefficient, evaluated against a binary fail-safe halt as a deployed-policy comparison (L7, Section 7).

**RQ1:** can the RSSM+ANFIS signal yield a graded risk coefficient that a controller can act on, where calibrated reactive baselines yield only alarms? **RQ2:** can fuzzy risk quantification give a less disruptive alternative to a fail-safe halt on recoverable disturbances? **RQ3:** can the whole loop run within a CPU-only real-time budget (<100 ms)?

This work makes three contributions. First, it integrates latent perception, RSSM prediction error, fuzzy risk quantification, and bounded residual control into one closed recovery loop. Second, it provides Webots-online validation under condition-based TRANSIT dynamics ($N = 40$ per mode per scenario) on four recoverable motion disturbances plus two out-of-scope semantic/actuation faults, where the ON policy improves RSR over fail-safe halt on the harder disturbances and improves trajectory smoothness on three of the four (Sections 6 and 7). Third, it reports a CPU-only P95 latency with substantial headroom under the real-time budget (Section 6). Section 2 surveys prior work, Sections 3 and 4 formalise the problem and architecture, Sections 5–7 cover evaluation and findings, and Section 8 closes the paper.

## 2 Related Work

Ten representative works (2023–2026) are compared along four functional modules: (M1) Perception, (M2) World Model, (M3) Risk Assessment, and (M4) Control, plus a fifth property, proportional (non-binary) recovery. The full coverage matrix and the criterion used for each column are given in the supplement. Four systems already cover all four modules (SafeDreamer [5], FARL [6], WM-AD [7], RAPT [8]), but none scales recovery proportionally to risk.

Residual RL [9] motivates bounded intervention but not semantic risk tiers, and Fuz-RL [10] applies fuzzy control without a perception front end. **Gap 1:** world models (DreamerV3 [2], V-JEPA 2 [1]) predict richly but lack a safety layer over prediction risk. **Gap 2:** no full 4/4-module system scales recovery continuously. WM-AD [7] halts at a threshold, FARL [6] switches policies binarily, and RAPT [8] delegates correction to a human operator. **Gap 3:** the detection literature (EfficientAD [11], FIPER [12], AHA [13]) flags failures but hands recovery to operators. This work closes these gaps with a graduated risk layer feeding a bounded residual controller.

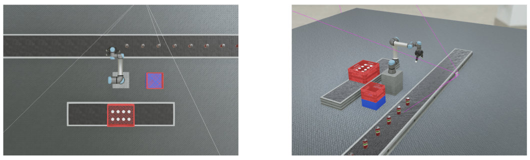


Fig. 1: Workspace layout (left) and mid-transit camera frame (right).

## 3 Problem Formulation

The task is modelled as a Markov decision process (MDP) [14], with state $s_t \in \mathcal{S}$, action $a_t \in \mathcal{A}$, transition $T(s_{t+1} \mid s_t, a_t)$, reward $R(s_t, a_t)$, and discount $\gamma$; the Webots cell instantiates this MDP. The hidden state contains robot joints and velocities, gripper state, object pose, task phase, and injected fault variables. The deployed agent sees only a 140×140 RGB crop $o_t \in \mathcal{O}$ through $\Omega(o_t \mid s_t)$, so the operational model is the POMDP $\langle \mathcal{S}, \mathcal{A}, \mathcal{O}, T, R, \Omega, \gamma \rangle$. Webots R2025a [15] advances at 125 Hz, and the recovery stack must keep CPU inference below 100 ms.

The nominal policy $\pi_{\text{nom}}$ executes the scripted task, while the recovery layer acts only in TRANSIT. For motion-level recoverable faults, a joint residual is dispatched, $a_t = \pi_{\text{nom}}(\hat{s}_t) + \alpha_t \delta a_t$ with $\delta a_t \in [-0.30, 0.30]^6$, where $\hat{s}_t$ is the runtime controller state (joints, gripper, phase, instrumented can pose, KL proxy, and clipped risk). The gripper command stays nominal, so premature release and wrong-product faults can be detected but not recovered with this residual layer.

The objective is graded intervention rather than a binary halt. During TRANSIT, DINOv2 maps $o_t$ to a latent token, the RSSM produces $e_t = D_{\text{KL}}(q \,\|\, p)$, and ANFIS combines the current KL with $x_{3,\text{std}}$ to assign $\alpha_t \in [0, 1]$ and {Safe, Medium, High, Critical}. The layer must identify anomalous risk, scale the correction by $\alpha_t$, preserve low-risk behaviour, and meet FPR $< 5\%$ and RSR $> 70\%$ on recoverable scenarios. This formulation assumes that DINOv2 features retain task structure [16], that RSSM error rises when observations become hard to predict [2], and that risk is monotone in the two ANFIS inputs [17].

## 4 Methodology

### 4.1 System Architecture

The pipeline is organised as four modules that move information from raw pixels to a graded corrective action within a single control timestep. At each step $t$, a frozen DINOv2 backbone encodes an RGB crop into a compact latent $z_t \in \mathbb{R}^{384}$, which then feeds a Recurrent State Space Model (RSSM) that emits a KL-divergence anomaly signal $e_t$. An ANFIS layer next fuzzifies this signal into a continuous risk coefficient $\alpha_t \in [0, 1]$, which finally modulates the magnitude of a bounded residual correction added to a nominal IK policy. The data flow is summarised in Fig. 2, and the workspace and camera viewpoint in Fig. 1.

### 4.2 Module 1: Visual Perception (DINOv2-ViT-Small)

Perception starts from a frozen DINOv2-ViT-Small encoder [16], exported to ONNX and executed through the OpenVINO provider for CPU inference. DI-

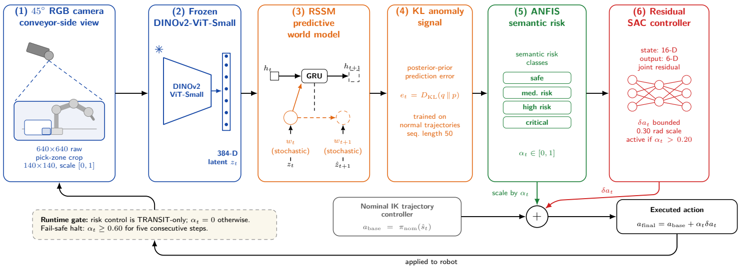



Fig. 2: Closed-loop pipeline: six functional blocks across four modules.

NOv2 is used because its self-supervised features supply task-relevant semantic structure without any anomaly labels, which matters given the limited simulation dataset. The encoder ingests an RGB crop from the 45-degree conveyor-side camera, resized to 140×140 and scaled to $[0, 1]$, and returns the final-layer CLS token $z_t \in \mathbb{R}^{384}$. Two choices keep this module real-time: the encoder is frozen to avoid overfitting while preserving its transferable semantics, and ViT-Small (21M parameters) is chosen as the largest backbone that fits the 100 ms CPU budget. The 140×140 crop carries $140^2/224^2 \approx 39\%$ of the pixel count of the standard 224×224 input; a 224×224+onnxruntime (CPU) prototype measured P95 = 150 ms during early development, already over budget; the deployed 140×140+OpenVINO configuration measures P95 = 34.24 ms (Section 6.4); resolution and execution provider both differ between these two figures, so they are not a controlled ablation. The trade-off is reduced sensitivity to fine gripper-state details.

### 4.3 Module 2: Predictive World Model (RSSM)

A latent encoder alone cannot indicate which observations are anomalous, so a model of the expected latent is also needed. This work therefore trains a Recurrent State Space Model (RSSM), following the latent-dynamics world model DreamerV3 [2], on the encoder's output during nominal trajectories. At deployment, the divergence between its posterior and prior is read off as the anomaly signal: when the world becomes hard to predict from the current hidden state, $e_t$ rises.

The RSSM combines a deterministic recurrent state $h_t$ (GRU, hidden size 128) with a stochastic state $w_t$ of dimension 32 governed by a prior $p(w_t \mid h_t)$. An encoder MLP [384 → 256] (LayerNorm + ELU) preprocesses $z_t$, and the posterior head concatenates $h_t$ with this representation to produce $q(w_t \mid h_t, z_t)$ (336,384 parameters total). The per-step anomaly signal is the closed-form Gaussian KL,

$$e_t = D_{\mathrm{KL}}\big(q(w_t \mid h_t, z_t) \,\|\, p(w_t \mid h_t)\big)\,, \tag{1}$$

which is small when posterior and prior agree and grows whenever the latent observation deviates from prediction. Training runs 50 epochs on 200 trajectories (Adam, lr $= 3\times10^{-4}$, sequence length 50, free-bits 0.5). Empirically, $e_t$ alone is informative but not decisive: per-episode maximum KL distributions for nominal

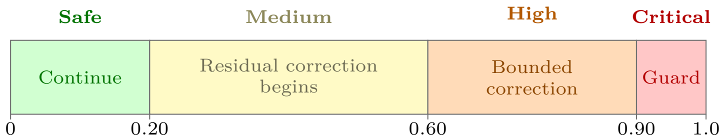
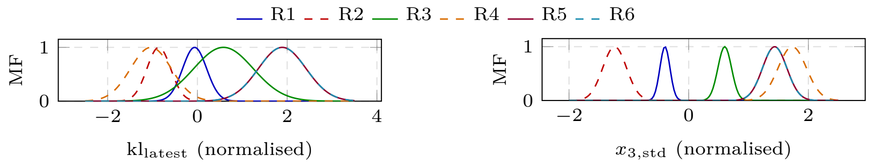



Fig. 3: Risk-tier action map over the risk coefficient $\alpha_t$. The residual controller dispatches for any $\alpha_t > 0.20$; 0.60 is where OFF's fail-safe halt arms, and 0.90/8 steps is ON's emergency guard (Section 4.5).




(a) kl$_{\text{latest}}$ input. (b) $x_{3,\text{std}}$ input.

Fig. 4: Learned Gaussian membership functions for the two ANFIS inputs (6 rules, normalised feature space).

and anomalous trajectories overlap (medians 0.31 vs. 0.39, with the B3 calibration threshold $0.4132 = 1.1\times$ the max calibration KL), so a single-threshold detector inherits a hard precision-recall trade-off. The next module is designed to resolve this ambiguity.

### 4.4 Module 3: Semantic Risk Quantification (ANFIS)

To turn the ambiguous $e_t$ into a controller-usable signal, it is routed through an Adaptive Neuro-Fuzzy Inference System (ANFIS) [3,4,17], which yields a continuous risk coefficient and an interpretable risk tier from learned membership functions. The deployed model takes two inputs, kl$_{\text{latest}}$ (the current KL) and $x_{3,\text{std}}$ (the 10-frame sliding std of the absolute pixel difference over the 140×140 crop after a 7×7 box blur, with no ROI mask). From these, a Sugeno-style network ($n_{\text{rules}} = 6$, sigmoid output) produces $\alpha_t \in [0, 1]$, which is mapped to the semantic risk tiers in Fig. 3. The Critical tier is exercised only as the ON-mode emergency backstop ($\alpha \geq 0.90$ for 8 steps), while the lower three tiers carry the active risk-aware control. The illustrative template *IF* KL is "Medium" *AND* $x_{3,\text{std}}$ is "Rising" *THEN* Risk is "High" shows the intended human-readable form. Because the target $\alpha^* = \text{clip}(\text{KL}/0.8 + x_{3,\text{std}}/0.05, 0, 1)$ is itself a clipped-linear combination, ANFIS is used strictly as *semantic-tier labelling infrastructure*, with no claim of a regression-fidelity improvement over a calibrated linear regressor (L5; see Section 7). This two-input network is fit to 10,000 procedurally-sampled points of the $\alpha^*$ surface (80/20 split) by minimising the logit-space MSE on targets clipped to $[0.01, 0.99]$, which the six-rule network reproduces closely. Fig. 4 shows the learned Gaussian membership functions.

### 4.5 Module 4: Risk-Aware Adaptive Control

The final module closes the loop. The intervention magnitude is designed to grow smoothly with estimated risk rather than switch abruptly at a threshold, implemented with a residual Soft Actor-Critic (SAC) [9,18] whose output is gated and scaled by $\alpha_t$. The policy observes a 16-dimensional state (joint positions, gripper state, can position, phase one-hot, KL proxy, and clipped $\alpha_t$) and returns a 6-dimensional joint residual. Recovery is intentionally joint-space only, so S5/S6 fall out of scope, and the `can position` is drawn from Webots ground-truth (L4). Whenever $\alpha_t > 0.20$, the policy emits $\delta a_t$, which is dispatched as

$$a_{\text{final}} = \pi_{\text{nom}}(\hat{s}_t) + \alpha_t \cdot \delta a_t, \tag{2}$$

with $\delta a_t$ bounded by a 0.30 rad action scale, making the correction proportional to risk. The training reward $R = R_{\text{task}} - 0.01 \cdot \|\alpha_t \cdot \delta a_t\|^2$ shares the same shape: the squared-residual penalty (scaled by $\alpha_t^2$) encourages near-zero intervention at low risk and tightens action only when risk warrants it. Evaluation compares two deployed policies on the same signal: ON (SAC enabled, with an emergency guard at $\alpha \geq 0.90$ for 8 steps) vs. a binary fail-safe halt (OFF, with SAC disabled and an abort at $\alpha \geq 0.60$ for 5 steps). This is a deployed-policy comparison, not an isolated residual-SAC ablation (L7).

Training runs in a lightweight offline kinematic simulator with procedurally sampled $(\mathrm{kl}_t, \alpha_t)$ pairs calibrated to Webots fault statistics; the deployed pipeline receives live RSSM output. Training runs for 1,500 episodes from a single seed 42 (L8): replay buffer 100,000, 1,024 warmup steps, Adam ($\mathrm{lr} = 3{\times}10^{-4}$), $\gamma = 0.99$, $\tau = 0.005$, batch 256, entropy target $-6$ nats. Anomaly perturbations are injected with probability 0.85 per episode under domain randomisation, and evaluation seeds are kept disjoint from training (L8 states what this does and does not imply for the reported Wilson 95% CIs).

*Condition-based TRANSIT.* TRANSIT is condition-based rather than fixed-budget: it terminates when the joint-position error from the reference pose stays below 0.08 rad for three consecutive steps, or at a 300-step cap (2.4 s at 125 Hz). At the end of an episode, the controller recycles the placed can and delivers the next supply over shared IPC; no invalid-supply cascade occurred across the 320 recoverable-scenario episodes. The end-to-end CPU-only latency is reported in Section 6.4.

# 5 Experimental Setup

## 5.1 Simulation Environment

All experiments run in Webots R2025a [15] with a Universal Robots UR5e (6-DOF) and Robotiq 3-finger gripper, under ODE physics with an 8 ms timestep (125 Hz). A 45-degree conveyor-side RGB camera (640×640 raw) provides observations, and the pick-zone crop is resized to 140×140 for the pipeline. The multi-controller architecture communicates over shared IPC files, which allows fault injection and state logging without touching the physics loop.

## 5.2 Failure Scenarios

Table 1 defines the six evaluated scenarios, spanning the temporal/kinematic (S1–S4) and logical (S5–S6) categories in a robotics-focused anomaly taxon-

Table 1: Failure scenarios and intervention scope.

| ID | Scenario | Injection | Breaks? | Recover? |
|---|---|---|---|---|
| S1 | Vibration | sinusoidal joint perturb. | no | yes |
| S2 | Jerky Motion | step-hold motion | no | yes |
| S3 | Pause | mid-transit freeze | no | yes |
| S4 | Speed Anomaly | reduced joint speed | no | yes |
| S5 | Premature Release | gripper opens mid-carry | yes | no |
| S6 | Wrong Product | wrong object supplied | yes | no |

omy [19]. The scenario selector is fixed at the start of an episode and each perturbation is applied through command-level hooks (joint timing/velocity, gripper state, or supplied object), leaving the Webots physics engine unchanged. The **Breaks?** column marks faults whose nominal task cannot complete without intervention beyond continuing the trajectory. Because residual correction is joint-space only, S5 and S6 fall outside its scope. RSSM and DINOv2 are trained *entirely unsupervised* on normal trajectories, so no labelled anomaly data enters training.

### 5.3 Episode Workflow and Phase Gating

Each episode runs an eleven-phase pick-and-place cycle in three stages: pick (home, approach, lower, close, lift), TRANSIT carry, and place (approach, lower, open, retract, home). Fault injection (S1–S6) and Module 4 intervention are confined to TRANSIT, where ANFIS computes $\alpha_t$ at every step and the residual-SAC layer dispatches bounded corrections whenever $\alpha_t > 0.20$. All other phases execute the nominal IK trajectory with $\alpha_t = 0$. TRANSIT terminates either by convergence (three consecutive steps with joint error $< 0.08$ rad) or at a 300-step cap. Within TRANSIT, $\alpha_t$ is damped with a 100-step confidence ramp ($\min(1, \text{transit step}/100)$, while KL $< 0.65$) to suppress transient high-KL artefacts during RSSM warm-up, and the fail-safe halt adds a 35-step grace period before it arms. The runtime ANFIS reads $\text{kl}_{\text{latest}}$ from the RSSM and $x_{3,\text{std}}$, both defined in Section 4.4.

### 5.4 Baselines and Ablations

Four diagnostic detector baselines plus one feature ablation are defined. **B1** is a Reactive Threshold on the DINOv2 embedding ($\ell_2$ distance to a nominal-mean embedding; no world model). **B2** is EfficientAD-style [11] (autoencoder reconstruction error on the same embedding). **B3** is RSSM KL-only (Module 2 only). **B4** is RSSM+ANFIS detect-only (Modules 1–3, no recovery). $x_{3,\mathbf{std}}$ **only** replaces the embedding with the raw pixel-motion feature from Module 3 (fixed score threshold, not FPR-calibrated; no embedding, no world model). B1/B2 are calibrated to 0.0% FPR on 10 nominal episodes (conservative, not sweep optima), and treated as diagnostic matched-calibration baselines in Section 6.3 because they lack a separately held-out normal set. B3 uses the threshold $0.4132 = 1.1\times$ the maximum calibration KL. B4 uses a fixed $\alpha \geq 0.92$/3-step threshold, calibrated independently of Ours' deployed thresholds ($\alpha \geq 0.60/5$ steps OFF, $\alpha \geq 0.90/8$ steps ON). B3/B4 FPR is evaluated on the remaining 70 nominal episodes. The ablation compares deployed intervention policies rather than a matched-threshold module-removal RSR.

### 5.5 Evaluation Metrics

**FPR** is the online false positive rate. **RSR** (Recovery Success Rate) excludes only invalid-supply events (none occur on recoverable scenarios). **Smoothness** reports the mean joint deviation from the TRANSIT reference pose (rad) and the mean residual magnitude $\|\alpha_t \delta a_t\|$ (rad) accumulated over TRANSIT. **TRANSIT outcomes** distinguish TRANSIT-reached (convergence) from TRANSIT-timed-out (300-step cap), and a timeout is not counted as a task failure when placement succeeds. Episode 0 is run and then excluded as warm-up, so OFF/ON share matched scenario configurations of $N = 40$ analysed episodes per mode per scenario, with conveyor supply and episode-synchronised recycle. Code, checkpoints and artifact paths are released with the accompanying supplementary material.

## 6 Results and Analysis

All results below are Webots-online measurements ($N = 40$ episodes per mode per scenario, with episode 0 excluded as warm-up) under episode-synchronised conveyor supply. Offline AUROC values are retained only as supplementary diagnostics and are not reported as operational outcomes.

### 6.1 Overall Performance

Table 2 pools the four recoverable scenarios (S1–S4, $N = 40$ each, denominator 160) into a single pooled RSR metric. Two deployed policies are compared on the same RSSM+ANFIS signal: OFF aborts at $\alpha \geq 0.60/5$ steps, ON continues with a bounded residual correction under an emergency guard at $\alpha \geq 0.90/8$ steps. The gap therefore reflects both the threshold relaxation and the corrective layer (L7, Section 7). On the pooled denominator, ON improves over OFF by +21.25 pp (138/160 vs. 104/160). B3 and B4 are detector-only diagnostics (recovery N/A; TPR in Table 3), with FPR evaluated over the held-out 70 nominal episodes (10 of the 80 used for B3 calibration). B1 and B2 are omitted here because their 0% FPR is a calibration target, not a held-out measurement. For Ours, the observed FPR is 0/80 on the full nominal set (Wilson CI $[0.0, 4.6]$). End-to-end CPU latency is P95 = 34.24 ms (mean 25.63 ms, P99 = 38.17 ms, see Section 6.4).

### 6.2 Per-Scenario Performance and Trajectory Smoothing

Per-scenario OFF/ON detail is given in Table 4 and plotted in the supplement. Gains concentrate on the harder disturbances: **S2 Jerky** (+25.0 pp) and **S4 Speed** (+42.5 pp) show non-overlapping Wilson 95% CIs, while S1 and S3 remain within CI overlap. Smoothing is complementary, with the joint-deviation

Table 2: Webots-online system summary (Wilson 95% CI in brackets). $N$ is the nominal-episode count: B3 and B4 are evaluated on the 70 episodes left after the 10 used to calibrate B3; Ours needs none and is evaluated on all 80. OFF's abort threshold is a fixed design choice, not evaluated on nominal episodes.

| Method | Role | FPR [95%CI] ($N$) | Pooled RSR [95%CI] |
|---|---|---|---|
| B3: RSSM KL-only | Detector | 0.0% $[0, 5.2]$ (70) | N/A |
| B4: RSSM+ANFIS | Detector+risk | 0.0% $[0, 5.2]$ (70) | N/A |
| Fail-safe halt (OFF) | Binary halt | N/A | 65.0% $[57.3, 72.0]$ |
| **Ours (ON)** | **Adaptive ctrl** | **0.0% $[0, 4.6]$ (80)** | **86.3% $[80.1, 90.7]$** |

Table 3: Detector TPR / intervention rate on S1–S4 (Wilson 95% CI).

| System | S1 Vib. | S2 Jerky | S3 Pause | S4 Speed |
|---|---|---|---|---|
| B1: Reactive Thresh. | 30.0 [18.1,45.4] | 20.0 [10.5,34.8] | 20.0 [10.5,34.8] | 22.5 [12.3,37.5] |
| B2: EfficientAD-style | 40.0 [26.3,55.4] | 32.5 [20.1,48.0] | 32.5 [20.1,48.0] | 35.0 [22.1,50.5] |
| B3: RSSM KL-only | 15.0 [7.1,29.1] | 32.5 [20.1,48.0] | 27.5 [16.1,42.8] | 17.5 [8.7,32.0] |
| B4: RSSM+ANFIS (detect) | 0.0 [0,8.8] | 0.0 [0,8.8] | 2.5 [0.4,12.9] | 0.0 [0,8.8] |
| $x_{3,\mathrm{std}}$ only | 0.0 [0,8.8] | 0.0 [0,8.8] | 0.0 [0,8.8] | 0.0 [0,8.8] |
| **Ours int.** ($\alpha > 0.20$) | **67.5 [52.0,79.9]** | **100 [91.2,100]** | **100 [91.2,100]** | **100 [91.2,100]** |

reduction ranging from 15.4% (S1) to 44.8% (S3). S4 is essentially unchanged because the velocity cap dominates the speed fault. The mean per-step residual EWR stays below 0.20 rad across all scenarios (clip 0.30 rad), and the ON-mode timeout fractions are highest on S2, S3 and S4 (revisited below).

### 6.3 Baselines and Feature Ablations

Table 3 reports detector TPR on the condition-based TRANSIT for S1–S4. B1/B2 are calibrated to 0% FPR on 10 normal episodes (diagnostic); B3 uses 0.4132 (1.1× max calibration KL); B4 uses $\alpha \geq 0.92$/3-step. **Ours** reports how often the controller engages ($\alpha > 0.20$), which is an intervention rate rather than an alarm TPR and is not comparable to the pooled RSR in Table 2. Three readings stand out. First, pixel-only $x_{3,\mathrm{std}}$ collapses to 0% TPR at this uncalibrated threshold, and the supplement obtains 0.6% TPR against a calibrated 0%-FPR threshold, so raw pixel-motion alone is insufficient here; the embedding-based detectors (B1/B2/B3) all reach non-zero TPR. Second, B1/B2/B3 reach moderate TPR (15–40%) but only as binary alarms, with no path to graduated correction. Third, tightening to B4's high-confidence criterion drops detector TPR to $\leq$ 2.5%, so at these conservative operating points thresholding supplies either low FPR or coverage, not both.

### 6.4 Consolidated Metrics and Latency

Table 4 consolidates the operational metrics for S1–S4: RSR (with Wilson 95% CI); the OFF/ON mean joint deviation $j$ (rad, success-conditioned) and its smoothing ratio Smooth $= (j_{\mathrm{OFF}} - j_{\mathrm{ON}})/j_{\mathrm{OFF}}$ (positive = smoother on ON); the ON-mode mean residual magnitude EWR $= \mathbb{E}[\|\alpha_t \delta a_t\|]$; and the ON-mode

Table 4: Per-scenario measurements ($N = 40$ per mode; symbols as defined in the text above; T-out is the ON-mode TRANSIT-timeout count). Pooled row aggregates S1–S4 ($N = 160$).

| **Scenario** | **RSR [CI]** | | **$j$ (rad)** | | **Smooth** | **EWR** | **T-out** |
|---|---|---|---|---|---|---|---|
| | OFF | ON | OFF | ON | | ON | ON |
| S1 Vibration | 70.0 [54.6,81.9] | 75.0 [59.8,85.8] | 0.757 | 0.640 | +15.4% | 0.086 | 6/40 |
| S2 Jerky | 67.5 [52.0,79.9] | **92.5 [80.1,97.4]** | 0.812 | 0.595 | +26.7% | 0.196 | 37/40 |
| S3 Pause | 65.0 [49.5,77.9] | 77.5 [62.5,87.7] | 0.847 | 0.468 | +44.8% | 0.176 | 25/40 |
| S4 Speed | 57.5 [42.2,71.5] | **100.0 [91.2,100]** | 0.928 | 0.941 | −1.4% | 0.199 | 40/40 |
| **Pooled (S1–S4)** | 65.0 [57.3,72.0] | **86.3 [80.1,90.7]** | 0.836 | 0.661 | +21.0% | 0.164 | 108/160 |

TRANSIT-timeout count, with Smooth computed from unrounded $j$. In the pooled row, RSR and T-out are episode-level over $N = 160$, whereas $j$ and EWR are unweighted means of the four per-scenario values. S5/S6 are omitted because both modes give 0% RSR (L1).

*TRANSIT-timeout reading.* Timeout is a phase-exit signal rather than a task-failure flag: placement still enters the RSR count. Mean TRANSIT steps (ON/OFF): S1 118/64, S2 300/158, S3 265/96, S4 300/300. OFF's fail-safe still fires on S4 (17/40 episodes, mean 77 steps); the 23 successes that remain all exhaust the cap, hence 300. Most S2 (37/40) and all S4 episodes reach the cap, so their RSR gains reflect task completion under a more permissive abort threshold (L7), not faster recovery (L9).

*Latency.* P95 = 34.24 ms (mean 25.63 ms, P99 = 38.17 ms) is measured on an i5-11320H CPU with OpenVINO, aggregated over 9,969 step-level samples from 19 deployed-controller sessions spanning seven scenario types (supplement). The real-time CPU budget ($< 100$ ms) is met with substantial headroom. The per-step cost also covers the condition-based TRANSIT bookkeeping.

## 7 Discussion

*RQ1: Online Risk Detection.* Pixel-level statistics alone are insufficient for detection: pixel-only $x_{3,\text{std}}$ collapses to 0% TPR (Table 3), while B1/B2/B3 all reach non-zero TPR on the same episodes. Conventional thresholds on these detector signals reach moderate TPR (15–40%) but only as binary alarms, and tightening to B4 ($\alpha \geq 0.92$/3-step) drops it to $\leq 2.5\%$: at these conservatively calibrated operating points, thresholding supplies either low FPR or coverage, not both. The RSSM signal is not the strongest binary detector (B2 matches or exceeds B3 on all four scenarios), but it alone yields a graded coefficient, so the controller engages at $\alpha > 0.20$ on the same signal and turns partial detection into graduated correction.

*RQ2: Risk-Aware Adaptive Control.* The ON policy improves RSR with non-overlapping Wilson 95% CIs on the harder disturbances (S2 Jerky +25.0 pp and S4 Speed +42.5 pp), while S1 (+5.0 pp) and S3 (+12.5 pp) remain within CI overlap. The OFF/ON gap couples threshold relaxation with residual correction

(L7), so the smoothing reduction on S1–S3 gives the cleaner controller-level view. The largest gains (S2, S4) also coincide with high ON-mode timeout fractions, so placement succeeds by absorbing the 300-step cap rather than by faster recovery. Beyond RSR, ON reduces mean joint deviation by 15.4% (S1), 26.7% (S2), and 44.8% (S3) at a mean per-step residual (EWR) below 0.20 rad, against a 0.30 rad clip. S4 is essentially unchanged because the velocity cap dominates.

*RQ3: Latency.* End-to-end P95 latency is 34.24 ms (9,969 step samples, 19 sessions), and the 100 ms budget is met with 65.8% headroom. The DINOv2 encoder, via ONNX/OpenVINO, dominates the runtime. Both targets from Section 3 are met: FPR is 0% against a 5% ceiling, and every per-scenario ON RSR clears 70%.

*Limitations.* **(L1)** Recovery is joint-space only: S5/S6 yield 0% RSR in both modes, since neither fault is expressible as a 0.30 rad joint residual. **(L2)** Neither the ANFIS layer nor the residual SAC (bounded by the 0.30 rad clip) is isolated by ablation. **(L3)** All numbers are from Webots, at wall-clock cost under a non-real-time simulation, and the 100 ms budget is a design target rather than a cell requirement; **(L4)** the SAC observation's Webots-instrumented can pose must become a perception-derived estimate for physical deployment. **(L5)** Because $\alpha^*$ is a clipped linear combination of KL and $x_{3,\mathrm{std}}$, ANFIS is used for semantic labelling, with no claim that it is superior in MSE to a calibrated linear regressor. **(L6)** High ON-mode TRANSIT-timeout counts mean placement success differs from fast recovery. **(L7)** The OFF/ON RSR gap couples threshold relaxation ($\alpha \geq 0.60/5 \rightarrow \alpha \geq 0.90/8$) with residual correction, so the residual contribution is not separately quantified, and the smoothing analysis is the closest proxy. **(L8)** A single training seed (seed 42) is used, and the residual policy is trained on procedurally sampled pairs offline but deployed on live RSSM output, so the Wilson 95% CIs capture neither training-seed variance nor this input-distribution shift. **(L9)** The RSR-vs-cycle-time trade on S2/S4 has not been quantified. **(L10)** No image-aligned per-frame KL/$\alpha$/residual traces are included in the main paper.

## 8 Conclusion

This work presented a four-module architecture for predictive anomaly detection and risk-aware adaptive control in pick-and-place. It combines a frozen DINOv2 encoder, an RSSM KL-divergence anomaly score, ANFIS graduated risk, and a bounded residual-SAC correction to turn one latent prediction signal into proportional intervention, and it addresses three gaps in prior work (world-model prediction without safety logic, recovery without graduated risk scaling, and detection without robot-side response). Webots-online evaluation under condition-based TRANSIT (Section 6) answers all three research questions, with the detection claim scoped as in Section 7 (RQ1). Future work targets L1–L10: a matched-threshold ablation, a multi-seed sweep, image-aligned per-frame traces, time-to-place, an ANFIS-vs-linear comparison, physical UR5e transfer, and S5/S6 recovery.

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